

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 6

Subject Code & Name: BP604T - BIOPHARMACEUTICS AND PHARMACOKINETICS

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

- | | |
|--|---|
| A) What is Area Under Curve? Enlist different methods used to calculate AUC. | 2 |
| B) Define loading and maintenance dose. | 2 |
| C) Differentiate between active transport and passive transport of drugs. | 2 |
| D) Define Bioavailability and Pharmacokinetics. | 2 |
| E) Enlist the components of blood to which drug binds. | 2 |
| F) List out different factors which affect renal excretion of drug. | 2 |
| G) What is meant by Compartmental Modeling? Classify them. | 2 |
| H) What is mean by Apparent volume of distribution? | 2 |
| I) Write the various factors influencing drug absorption through GIT. | 2 |
| J) Explain BCS classification of drug. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

- | | |
|--|----|
| A) Define Non-linear Pharmacokinetics. Explain methods to determine non linearity and causes of non-linearity. | 10 |
| B) Define drug absorption. Explain various mechanisms of drug absorption through GIT. | 10 |
| C) Discuss in detail one compartment open model for a drug administered by IV bolus. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

- | | |
|--|---|
| A) Explain Michaelis-Menten equation and determine of Vmax and Km. | 5 |
| B) Explain drug plasma-level time curve along with its Pharmacokinetics parameter. | 5 |
| C) Explain Mammillary and Catenary Model. | 5 |
| D) Describe about the physiological barrier to the distribution of drug. | 5 |
| E) Explain Phase-I reactions of drug Metabolism. | 5 |
| F) Write a note on non-renal excretion of drugs. | 5 |
| G) Explain the factors affecting protein binding of drug. | 5 |
| H) Write a note on In-Vitro and In-Vivo Co-relation. | 5 |
| I) List USP In vitro dissolution test apparatus and illustrate dissolution Apparatus 1 and Apparatus 2 as per USP. | 5 |

*** * * End * * ***